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COMPONENTIAL ANALYSIS AND THE STUDY OF MEANING

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THE PROBLEM

1. Introduction. That the methods of componential analysis as they have been developed for analyzing linguistic forms are applicable in principle for analyzing other types of cultural forms is a proposition toward whose demonstration I have for some time sought to orient my ethnographic researches. The results of some exploratory work toward this end have already been published.[1] Included among them is an analysis of Truk kinship terminology, in which it proved possible to apply some of the principles of linguistic analysis to the problem of deriving the significata[2] of kinship terms and of determining which terms went together in what I called semantic systems. I am taking up this material again in order to present a fuller discussion of the method and of its implications for developing an empirical science of meaning.[3]

The aspect of meaning to be dealt with is signification as distinct from connotation. What is meant by these terms will become clear in the course of the discussion. Suffice it to say at this point that the significatum of a linguistic form is composed of those abstracted contextual elements with which it is in perfect association, without which it cannot properly occur. Its connotata are the contextual elements with which it is frequently but less than perfectly associated. Significata are prerequisites while connotata are probabilities and possibilities. Only the former have definitive value.

2. Methodological orientation. The problem of determining what a linguistic form signifies is very well illustrated by kinship terms. In essence it is this: what do I have to know about A and B in order to say that A is B's cousin? Clearly, people have certain criteria in mind by which they make the judgment that A is or is not B's cousin. What the expression *his cousin* signifies is the particular set of criteria by which this judgment is made.

This is analogous to the problem of determining what are the acoustical criteria which differentiate *sick* from *thick* so that we hear them as different linguistic forms instead of one form, as might a native speaker of German. In this case the criteria are a set of acoustical percepts which in varying combinations make up the phonemes of a language. A linguist arrives at a statement of

[1] W. H. Goodenough, *Property, kin, and community on Truk* (*Yale University publications in anthropology*, No. 46; 1951), hereafter *PKC*.

[2] *Significatum* and *denotatum* are used as defined by Charles Morris, *Signs, language and behavior* 17 (1946).

[3] I wish to thank Henry M. Hoenigswald for his encouragement and John Cole for many fruitful discussions. Field work on Truk was undertaken in 1947 in connection with the Yale University expedition under George P. Murdock, part of the Coordinated Investigation of Micronesian Anthropology sponsored by the Pacific Science Board of the National Research Council, financed by the Office of Naval Research, the Wenner-Gren Foundation for Anthropological Research, and the Department of Anthropology of Yale University.

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them not by direct observation, but by testing various hypotheses which he formulates about them until he finds a hypothesis which fits the acoustical phenomena as he has noted them and which also provides him with a model for producing acoustical phenomena himself which result in predictable responses by others. The first step in this procedure is, of course, to record as many discriminable differences in the acoustical phenomena as possible by means of a phonetic system of notation. Not all discriminable differences, however, serve as criteria by which functionally distinct categories of sound, phonemes, are differentiated. It is an object of linguistic analysis by systematically examining the mutual distributions (in recorded speech) of the acoustical phenomena as phonetically noted, to produce the most adequate possible theory as to what are the language's phonemes, its elementary phonological components.

Now let us suppose that the language under study is a written one, and that the notation used by those who are literate in it is partially phonemic, but not perfectly so. There are some phonemes which are written with more than one symbol and some which are written with the same symbol, e.g. the identical phonemes of English *see* and *sea* and the different phonemes of English *read* in the expressions *will read* and *have read*. Let us suppose, furthermore, that it is the linguist's job not only to determine what the phonemes of the language are but to show how they relate to the symbols conventionally used for writing it. To do this he would have to get a literate speaker to read him a text written in the conventional alphabet. He would have to record this text, as it was read, in a phonetic notation and derive the phonemes in the prescribed manner. He would then have to draw up a phonemic transcription of the text, compare it with the text as written conventionally, and compare both with the text as recorded phonetically in order to make a precise statement of what are the phonological elements in the language for which the conventional symbols stand.

It is the situation just described which is analogous to the one facing the semantic analyst. While he aims to find the conceptual units out of which the meanings of linguistic utterances are built, he has the conventional symbols of speech which more or less stand for these units (or combinations of them) already given him. He must acquire an informant who knows how to use these symbols. The procedure is to note what speech symbols the informant uses in what contexts, and at the same time to describe these contexts by means of a notation which makes as many discriminations as conveniently possible. Such a notation is analogous to the phonetic notation of the linguist. In connection with kinship terminology, for example, we note that the contextual elements in connection with which an English-speaking informant uses the expression *my cousin* are among other things capable of description variously as FaBrSo (father's brother's son), FaSiSo, FaBrDa, FaSiDa, FaFaSiSo, FaMoMoBrSoDa, etc. The distinctions made in this notation are not the same as those which distinguish between the cultural categories of kin signified by kinship terms, just as the distinctions of phonetic notation are not the same as those between the cultural categories of speech sound which linguists call phonemes. What the notation describes is the semantic equivalent of the allophones of phonemic analysis—what, if we

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draw upon Bloomfield's terminology, we may call allosemes, i.e. denotative manifestations of sememes.[4] By examining the mutual arrangements or distributions of allosemes (denotative types) obtained in this fashion we can arrive at a statement of sememes (significative types). Then we are in a position to see how these relate to speech symbols and, in doing so, to see what speech symbols are homonyms (stand for more than one sememe or have more than one significatum) and what pairs of symbols are synonyms (stand for the same sememe or have the same significatum).[5]

As organizations of acoustical phenomena, phonemes are analyzable as combinations of percepts which we conventionally describe with reference to the manner of their production in speech. Thus one set of percept values relates to place of articulation (labial, alveolar, palatal, etc.), another to voicing, and a third to mode of articulation (stopped, nasalized, spirantic, lateral, etc.). If we regard these as three distinct variables each with a set of modal values characteristic for a given language, then any phoneme of the language can be described as a particular combination of these values (among others). The minimum number of such variables and values necessary to account for the differences between the language's phonemes—the variable of voicing is irrelevant for Trukese, for example—constitute the perceptual (acoustical) components of its phonology.

In a similar manner, as we shall see, sememes consist of combinations of percepts and/or concepts, which consist of values for different variables. Just as we may say that the English phonemes /t/ and /d/ complement each other with respect to the variable of voicing, or /p/, /t/, and /k/ are a complementary set with respect to place of articulation, it is possible for us to arrange sememes into complementary sets, e.g. the sememes of *old* and *young* complement each other with respect to a conceptual variable of age, *come* and *came* with respect to tense, *husband* and *wife* with respect to sex.

Any set of linguistic forms, whatever their shape, which signify complementary sememes may be said to belong to the same PARADIGM. The paradigms of conventional Latin grammar, to take a familiar example, consist of forms whose semantic values are expressions of such conceptual variables as person, number, tense, etc. Such paradigms are easy to construct through inspection only, because the linguistic forms involved usually have obvious morphological similarities, e.g. *amō, amās, amat; agricola, agricolae, agricolam.* Morphologically obvious para-

[4] Leonard Bloomfield, *Language* 162 (1933). Just as a phone is a material (behavioral) manifestation of a phoneme, a denotatum is a manifestation of a significatum. For different applications of Bloomfield's terminology see Eugene A. Nida, *Word* 7.1–14 (1951), and Joseph H. Greenberg, *Language in culture* 3–19 (ed. Harry Hoijer; 1954). As will appear more plainly later, I use the term *sememe* in a somewhat different sense from Bloomfield.

[5] I would carry the analysis further than Nida (8–9), who does not differentiate signification from connotation and metaphorical usage. By staying with linguistic forms and treating meanings as if they were self-evident, he fails to come to grips with the semantic problems. He turns his back on the 'common denominator' approach because he has no analytical method for going beyond a simple listing of allosemes. Any speaker of a language is able to use a given form in new contexts in ways perfectly intelligible to other speakers. Analysis must enable us to do the same.

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digms, however, constitute a limited special case. That significata or sememes generally form paradigmatic structures, and that their analysis can in turn tell us much about human cognitive processes are things which cannot be so readily appreciated without recourse to some such operationally systematic approach to semantics as will be illustrated here.[6]

To demonstrate the method, we shall analyze the kinship terminology of Truk. It is well suited for this purpose, because, while fairly straightforward, it exemplifies several noteworthy things about the relation between linguistic forms and their significata.

The Method: Truk Kinship Terms

3. Determining the universe. The first step in analysis is to gather together all expressions whose denotata make it appear on inspection that there may be some common element in their significata; which is another way of saying that they appear to relate to the same general subject matter. In the present instance, for example, we are concerned with the subject matter of kinship in descriptive or referential usage (as distinct from vocative usage). We set about, therefore, collecting as many expressions as possible whose denotata suggest to us that they may belong to the universe of kinship, i.e. signify partitions of it.[7]

There are two procedures which we can follow. One approach is to start with an expression such that a sample of its denotata seems most clearly to put it in the realm of kinship. Any other expression whose denotata suggest that it complements the first in some way must, by virtue of complementation, relate to another partition of the same universe of which the first is also a partition. An expression, moreover, whose denotata are entirely included within the denotata of another expression or within the denotata of a complementary set of expressions, must also signify a partition of the same universe to which the latter expressions belong. Another approach is to start with an expression whose denotata appear to cover the entire universe in question and which appears, therefore, to signify something that we would translate, in this case, as 'kinsman'. All expressions whose denotata are entirely included within the denotata of this expression will belong to the universe of kinship. Where both approaches are possible, as with Truk, one serves as a check on the other.

In order to minimize problems arising from metaphorical usage, it is advisable to collect one's information systematically in a context in which the informant is being asked to give statements of fact according to his understanding and usage. For kinship, the genealogical method of collecting information, as de-

[6] Nida writes (6): 'A seme may be defined as (1) the meaning in a particular type of context of (a) a morpheme or (b) a formal part of a morpheme, or (2) a meaning implicit in the forms of a paradigmatic series. Semes of type 1 are overtly symbolized and those of type 2 are covertly indicated.' We shall see that his type-1 semes are combinations of his type-2 semes. The former are what I call sememes, the latter what I regard as the basic components of signification.

[7] By kinship I mean a series of Trukese social distinctions, and the terms signifying them, which more closely fit the cross-cultural concept of kinship than any other series of distinctions known in Truk.

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veloped by Rivers,[8] provides a systematic approach. It is always possible to check with an informant, moreover, whether the persons in question are properly or 'literally' in the relationship attributed to them in some text or conversation which has been recorded (again in his sense of the proper or literal).

By following these procedures for Truk, we obtain a large series of expressions, samples of whose denotata lead us to believe that they signify some part of the kinship universe. With respect to linguistic morphology these expressions are highly varied. Since we have collected all of them in the context of referential usage, they all occur with 'possessive' formations. In other respects their forms vary as follows:[9]

(a) a construction consisting of one root morph plus a suffixed possessive pronoun, e.g. *seme-j*, *pwii-j*, *neji* (a zero suffix here for first person singular);

(b) a construction consisting of a head of more than one root morph plus a suffixed possessive pronoun, e.g. *seme-nape-j*, *jine-ji-seme-j*;

(c) a phrase consisting of a construction of type a or b followed by a modifier, e.g. *jää-j mwään*, *neji feefin*;

(d) a phrase consisting of a chain of constructions of types a, b, or c connected by a linking form *-n* which makes that which immediately follows a possessor of that which precedes, e.g. *seme-n seme-j*, *nëwy-n pwii-n seme-j*, *jää-n seme-j mwään*, *nëwy-n pwii-j feefin*.

Some of these expressions, especially those of type d, will, we suspect, have significata which can be predicted from the significata and arrangement of their constituent forms. Thus *seme-n seme-j* denotes anyone who is *seme-j* to anyone who is *seme-j* to me and is analogous to the English phrase *my uncle's uncle*. Such expressions do not have to be kept in one's final description of meanings.[10] If we may anticipate results at this point, all expressions of type d and some of type c turn out to have significata which can be deduced from the significata of their parts, while others of type c and all those of type b lack deducible meanings. The latter must be treated as semantically equivalent, therefore, to expressions of type a, though differing morphologically.[11]

Any utterance whose signification does not follow from the signification and arrangement of its parts we shall hereinafter call a *lexeme*.[12] We shall refine this definition later. Meanwhile, it is with the limited number of lexemes pertaining

[8] W. H. R. Rivers, *The Todas* (1906).

[9] All kinship expressions are cited in the first-person singular form. Some morphemes which signify kinship with possessives do not do so without them; other kinship morphemes occur only as bound forms in possessive formations. The orthography is explained in *PKC* 26–8.

[10] This is true, of course, only with respect to descriptions of significata. *My uncle's uncle*, for example, may or may not be a kinsman of mine. The connotation, however, is that he is not; for the expression rarely occurs where the expression *my great-uncle* can be used.

[11] For further comment on the fact that expressions on similar morphological levels may belong to different semantic levels, see *Lg.* 31.244–5 (1955).

[12] Though I seem to use the term differently, my use of *lexeme* in fact corresponds quite closely to that of Eleanor H. Jorden, *The syntax of modern colloquial Japanese* 8–26 (*Lg. dissertation* No. 52, 1955). The main difference is that I have explicitly added significational criteria to the definition.

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LEXEMES	SAMPLES OF DENOTATA
semej (*sama, *ji)	Fa, FaBr, MoBr, FaFa, MoFa, FaFaBr, FaMoBr, MoFaBr, MoMoBr, FaSiSo, FaSiDaSo, SpFa, SpMoBr, SpFaBr, SpFaSiSo, MoSiHu, FaSiHu, etc.
jinej (*jina, *ji)	Mo, MoSi, FaSi, MoMo, FaMo, FaFaSi, FaMoSi, MoFaSi, MoMoSi, FaSiDa, FaSiDaDa, SpMo, SpMoSi, SpFaSi, SpFaSiDa, FaBrWi, MoBrWi, etc.
semenapej (*sama, *napa, *ji)	Fa, FaFa, MoFa.
jinenapej (*jina, *napa, *ji)	Mo, FaMo, MoMo.
jinejisemej (*jina, *ji, *sama, *ji)	FaSi, FaSiDa, FaSiDaDa, FaMo, FaMoSi, FaMoMo, etc.
pwiij (*pwii, *ji)	For male ego: Br, MoSiSo, FaBrSo, FaMoBrSo, FaSiSoSo, WiSiHu, etc. — For female ego: Si, MoSiDa, FaBrDa, FaMoBrDa, FaSiSoDa, HuBrWi, etc.
feefinej (*feefina, *ji)	For male ego: Si, FaBrDa, MoSiDa, FaMoBrDa, FaSiSoDa, but NOT WiBrWi. — For female ego: no denotata.
mwääni (*mwääni, *ji)	For male ego: no denotata. — For female ego: Br, MoSiSo, FaBrSo, FaMoBrSo, FaSiSoSo, but NOT HuSiHu.
mwëgejej (*mwëgeja, *ji)	For male ego: same as feefinej. — For female ego: same as mwääni.
jëësej (*jëësa, *ji)	For male ego: SiHu, WiBr, FaBrDaHu, etc. — For female ego: BrWi, HuSi, FaBrSoWi, etc.
pwynywej (*pwynywa, *ji)	For male ego: Wi, WiSi, BrWi, FaBrSoWi, etc. — For female ego: Hu, HuBr, SiHu, FaBrDaHu, etc.
jääj mwään (*jaa, *ji, *mwääni)	For male ego: o.Br, o.MoSiSo, MoBr, MoMoBr. — For female ego: o.Si, o.MoSiDa.
mwäänинyki (*mwääni, *nyky, *ji)	For male ego: y.Br, y.MoSiSo, SiSo. — For female ego: y.Si, y.MoSiDa.
neji (*nëwy, *ji)	So, Da, ChCh, BrCh, SiCh, MoBrCh, MoMoBrCh, FaBrChCh, MoSiChCh, FaSiSoChCh, FaSiDaSoChCh, FaMoMoBrChCh, etc.

TABLE 1

to kinship that our analysis will proceed. They are listed, together with their constituent morphemes (*) and samples of their possible denotata in Table 1.

From the sample of denotata collected for each of these lexemes it would certainly appear that they all relate to kinship and are therefore parts of the same semantic universe. We can confirm this impression from the lexeme *tefej*, which can denote anything denoted by any of the lexemes listed in Table 1 and does not denote anything that they fail to denote, except that it may also denote WiBrWi and HuSiHu. While the latter are among the possible denotata of *tefej*, they cannot be denoted by any of the other lexemes listed. Since we have no record of any other expression whose denotata are more nearly congruent with the aggregate of denotata for all the lexemes in Table 1, we may conclude that *tefej* signifies the universe in question, that the lexemes listed signify segments or partitions of it, and that one segment of it (WiBrWi and HuSiHu) remains which is not signified by any expression so far listed.

Investigation with informants shows that the gap just noted is filled only by a phrase of type d above, *pwynywe-n ëëse-j* (someone who is *pwynywej* to one who

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is *jëësej* to me). Since this usage is quite consistent with the possible denotata of its parts when thus combined, the expression belongs with the host of other expressions excluded from consideration on the ground that they can be freely and accurately generated by combining the expressions listed. In accordance with the definition of a lexeme given above, we can say that there is no lexeme in Trukese signifying a segment of the universe signified by *tefej* which includes WiBrWi or HuSiHu among its denotata. We shall have more to say about gaps of this kind in connection with paradigms below.

Since all possible denotata of *tefej* together define the extent of the universe of kinship for Truk, it will be well to describe them fully. In addition to all lineal ascendants and descendants of ego or his spouse, they include (a) members of ego's descent line, lineage, ramage, and subsib;[13] (b) members of ego's father's descent line, lineage, ramage, and subsib; (c) member's of ego's mother's father's descent line and lineage; (d) children of all men of a and b; (e) children of all d as long as their parents live; (f) all spouses of all a, b, c, d, and e; (g) all a, b, c, d, and e of ego's spouse; and (h) spouses of members of ego's spouse's descent line and lineage (see Figure 1 below). All such persons may be referred to as *tefej*, or by the appropriate lexeme from Table 1. Persons to whom a kinship bond can be traced beyond these boundaries are not *tefej* but *määrääri*, provided the connection has been mutually acknowledged and its obligations are mutually agreed to. Such persons are not referred to by any of the lexemes in Table 1.

It is clear from the foregoing description of the universe signified by *tefej*, that membership in or connection through ego's, ego's father's, or ego's spouse's matrilineal kin groups is a key consideration. This suggests that these groups may serve as boundaries for the various segments of the universe, i.e. as criteria for differentiating between kinsmen.

4. Distributions of lexemes in the universe of possible denotata. If we turn now to the kinship lexemes listed, we find that we can group them in various ways according to their denotata. Perhaps the most obvious basis for grouping them is in relation to sex differences. They give us three groups of lexemes, as follows.

Group 1: lexemes whose denotata are all of one sex only, regardless of ego's sex: *semej*, *jinej*, *semenapej*, *jinenapej*, and *jinejisemej*.

Group 2: lexemes whose denotata are all either of the same sex as ego or of the opposite sex, but never both: *pwiij*, *feefinej*, *mwääni*, *mwëgejej*, *jëësej*, *pwynywej*, *jääj mwään*, and *mwäänинyki*.

Group 3: lexeme whose denotata are of either sex, regardless of ego's sex: *neji*.

4.1. Group 1. Among immediate kin, all denotata are of a higher generation. Among remoter kin, all denotata not of a higher generation are in ego's father's matrilineal kin groups. This suggests the proposition that FATHER'S MATRILINEAL KIN GROUPS ARE INCLUDED IN HIGHER GENERATIONS.

In this group, the lexemes *semej* and *jinej* cover all possible denotata. Their denotata, moreover, do not overlap at any point. They are, therefore, a complementary pair, and it is evident that their denotata differ with respect to a variable

[13] All these Trukese matrilineal groups are described in *PKC*.

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	A	B	C	D	E
Higher Generations	NEJI *neji* neji *neji*	PWIIJ *pwynywej* MWËGEJEJ *jëësej* pwynywej *pwiij* jëësej ***	SEMEJ *jinej* JINEJ *semej* semej *jinej* jinej *semej*	SEMEJ *jinej* JINEJ *semej* semej *jinej* jinej *semej*	NEJI *neji* neji *neji*
Ego's Generation	NEJI *neji* neji *neji*	PWIIJ *pwynywej* MWËGEJEJ *jëësej* pwynywej *pwiij* jëësej ***	SEMEJ *jinej* JINEJ *semej* semej *jinej* jinej *semej*	PWIIJ *pwynywej* MWËGEJEJ *jëësej* pwynywej *pwiij* jëësej ***	NEJI *neji* neji *neji*
Lower Generations	NEJI *neji* neji *neji*	PWIIJ *pwynywej* MWËGEJEJ *jëësej* pwynywej *pwiij* jëësej ***	SEMEJ *jinej* JINEJ *semej* semej *jinej* jinej *semej*	NEJI *neji* neji *neji*	NEJI *neji* neji *neji*

FIGURE 1

Column A: descendants of children of men of ego's (or ego's spouse's) father's groups.
Column B: children of men of ego's (or ego's spouse's) father's groups.
Column C: members of ego's (or ego's spouse's) father's groups.
Column D: members of ego's (or ego's spouse's) groups.
Column E: descendants of men of ego's (or ego's spouse's) groups.

Generations are in accord with the strictly genealogical model.

Small capitals denote consanguineal kin of ego; italics denote their spouses.
Roman lower case denotes consanguineal kin of ego's spouse; italics denote their spouses.
Asterisks denote the 'zero lexeme'.

of SEX OF THE RELATIVE.[14] Their respective significata are, therefore, apparent functions of this variable (among others).

The denotata of *semenapej* are included in those of *semej*, and the denotata of *jinenapej* in those of *jinej*. Both *semenapej* and *jinenapej* complement each other with respect to the sex of the relative. Their denotata, moreover, include only lineal ascendants of ego, which implies the complementary conceptual categories consisting of those of one's *semej* and *jinej* who are not lineal ascendants, concepts which are not represented by any lexemes recorded for the universe of kinship.

The denotata of *jinejisemej* include only those of one's *jinej* who belong to one's father's matrilineal kin groups. There is no complementary lexeme for those of one's *semej* who belong to these kin groups, nor is there one for those of one's *jinej* who do not belong to these kin groups.

We may state the criteria relative to ego for the denotata of each of these lexemes (define their significata) as follows: *semej*: simultaneously (a) *tefej*, (b) of higher generation (including member's of father's matrilineal groups), and (c) male; *jinej*: simultaneously (a) *tefej*, (b) of higher generation, (c) female;

[14] If $a + b$ denote the class of all things that are either in a or in b but not in both, and if $a + b = 1$ (where 1 is any class of which a and b are subclasses), then a is the complement of b.

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semenapej: simultaneously (a) *semej* and (b) lineally connected; *jinenapej*: simultaneously (a) *jinej* and (b) lineally connected; *jinejisemej*: simultaneously (a) *jinej* and (b) of ego's father's matrilineal kin groups.

4.2. Group 2. The lexemes in this group fall into two sets; in one, relative age is not an apparent factor (*pwiij*, *feefinej*, *mwääni*, *mwëgejej*, *jëësej*, and *pwynywej*), in the other it is (*jääj mwään* and *mwääninyki*). If there were a lexeme representing the residual denotata of *tefej*, WiBrWi and HuSiHu, it would belong to the first set. We shall add this 'zero lexeme' to it for analysis.

While the lexemes within each set are complementary, the two sets as such do not complement each other and must be treated separately.

4.21. The denotata of all lexemes in the first set (*pwiij* etc.) are in ego's generation or have fathers who belong to the same matrilineal kin groups with ego's father. We have already stated that the father's matrilineal kin groups are included in higher generations. We now find support for this proposition in its corollary: THE CHILDREN OF MEN OF ONE'S FATHER'S MATRILINEAL GROUPS ARE INCLUDED IN ONE'S OWN GENERATION.

In order to simplify further analysis of this subset of kinship lexemes we shall use the following notation: α will represent all possible denotata of *pwiij*, β of *feefinej*, γ of *mwääni*, δ of *mwëgejej*, ϵ of *jëësej*, ζ of *pwynywej*, and η of the 'zero lexeme'.

If we let 1 stand for the universe represented by all possible denotata of all lexemes in this set, it is evident that

$$\alpha + \beta + \gamma + \epsilon + \zeta + \eta = 1$$

and also that $\alpha + \delta + \epsilon + \zeta + \eta = 1$

wherefore $\beta + \gamma = \delta$

Thus the possible denotata of *feefinej* and *mwääni* are complementary subsets of the possible denotata of *mwëgejej*; the significata of the former pair are discrete partitions of the latter; and *feefinej* and *mwääni* together are synonymous with *mwëgejej*.[15] Everything that will be said regarding the signification of *mwëgejej* will therefore be true of *feefinej* and *mwääni* considered as a unit.

If we set $A = \alpha + \delta + \eta$, and $B = \epsilon + \zeta$, every one of A is a member of ego's matrilineal kin groups, a child of a man of ego's father's matrilineal kin groups, or a spouse of a member of ego's spouse's matrilineal kin groups;[16] while every one of B is a member of ego's spouse's matrilineal kin groups, a spouse of a member of ego's matrilineal groups, or a spouse of a child of a man of ego's father's matrilineal groups (see Figure 1 below). Thus all A and all B are in complementary distribution, but the criterion differentiating them is obscure; for while there are no consanguineal kin in B, there are affinal kin in both A and B. If we bear in mind, however, the three sets of matrilineal groups (ego's, ego's father's, and ego's spouse's) through which kinship is traced, it becomes evident that

[15] When $a = b$ in the sense that any denotatum of a is also a denotatum of b, and conversely, then a and b are synonyms.

[16] These three clusters are known respectively as *pwiipwiicëk* 'just *pwii*', *pwiipwi winisam* '*pwii* with *semej* in common', and *pwiipwi winipwyny* '*pwii* with *pwynywej* in common'.

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any kinsman who has the same tie as ego to the kin group through which he and ego reckon their relationship, is a member of A; while any kinsman whose tie to the connecting group is different from ego's is a member of B. Kinsmen of the same generation in a symmetrical or parallel relationship to the connecting matrilineal group will refer to each other as *pwiij*, *mwëgejej*, or 'zero lexeme'; while those in an asymmetrical or nonparallel relationship will refer to each other as *jëësej* or *pwynywej*. The consistent difference between all A and all B reflects a variable of PARALLEL CONNECTION WITH INTERVENING KIN GROUPS.

If we set C = $\alpha + \epsilon$ and D = $\delta + \zeta + \eta$, the difference between all C and all D reflects a variable of SEX RELATIVE TO EGO'S SEX, all C being of the same sex as ego and all D of opposite sex.

Still to be determined is the difference between δ and η. It appears that all δ are consanguineally related to ego while all η are related otherwise, so that the difference between them reflects a variable of the CONSANGUINEAL/AFFINAL MODE OF RELATIONSHIP.

As for β and γ, the two subsets of δ, the difference between them may be described in terms of a variable of the SEX OF EGO or the SEX OF THE RELATIVE; it is immaterial which we choose.

We now state the criteria (relative to ego) for the denotata of each of these lexemes as follows: *pwiij*: simultaneously (a) *tefej*, (b) of the same generation, (c) of the same sex, and (d) with parallel relationship to the connecting kin group; *jëësej*: simultaneously (a) *tefej*, (b) of the same generation, (c) of the same sex, and (d) with nonparallel relationship to the connecting kin group; *mwëgejej*: simultaneously (a) *tefej*, (b) of the same generation, (c) of opposite sex, (d) with parallel relationship to the connecting kin group, and (e) consanguineal; *feefinej*: simultaneously (a) *mwëgejej* and (b) female; *mwääni*: simultaneously (a) *mwëgejej* and (b) male; 'zero lexeme': simultaneously (a) *tefej*, (b) of the same generation, (c) of opposite sex, (d) with parallel relationship to the connecting kin group, and (e) affinal; *pwynywej*: simultaneously (a) *tefej*, (b) of the same generation, (c) of opposite sex, (d) with nonparallel relationship to the connecting kin group.

4.22. The denotata of the two lexemes comprising the second subset of Group 2 (*jääj mwään* and *mwääninyki*) include only members of ego's matrilineal kin groups; there is no complementary lexeme for kinsmen not members of these groups. The denotata of these lexemes, moreover, comprise only persons who are the same sex as ego; but they do not complement the lexemes denoting kinsmen of opposite sex in ego's generation, because *jääj mwään* and *mwääninyki* together denote any kinsman of ego's sex in ego's matrilineal kin groups regardless of generation. The consistent difference between the denotata of the two lexemes reflects a variable of AGE RELATIVE TO EGO'S AGE.

We state the criteria relative to ego for the denotata of each of these lexemes as follows: *jääj mwään*: simultaneously (a) *tefej*, (b) of the same matrilineal kin group, (c) of same sex, and (d) older; *mwääninyki*: simultaneously (a) *tefej*, (b) of the same matrilineal kin group, (c) of same sex, and (d) younger.

Because generation is not a criterion for denotata of these lexemes, they do not complement any of the other lexemes in Table 1, with the exception of

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jinejisemej, which can denote only members of ego's father's matrilineal kin groups.

4.3. Group 3. The denotata of *neji*, the only lexeme in this group, include persons in lower generations than ego's, excepting persons in ego's father's matrilineal groups and children of men in these matrilineal groups. They also include the children of any men in ego's matrilineal kin groups together with their children, and the children of any children of men of ego's father's matrilineal groups (see Figure 1). Thus *neji* complements all the other lexemes discussed (except *jääj mwään* and *mwäänin*yki), the difference between its denotata and those of the others being dependent on a variable which, for lack of a better term, we call GENERATION HEIGHT.

We state the criteria for the denotata of *neji* as being for ego simultaneously (a) *tefej* and (b) of junior generation.

5. The concept of generation. It is obvious that what we are dealing with as generations do not coincide with what we would expect from the usual model of a genealogical table. Some modification of this model is needed to clarify this criterion in Trukese usage. To see the pattern involved, we plot the distribution of possible denotata for the several lexemes in Figure 1. Examination of this distribution enables us to formulate principles of generation equivalence in Truk: All members of ego's matrilineal kin groups belong to generations according to the genealogical model; for all other kinsmen the genealogical model applies as far as the following principles permit: (1) spouses are always in equivalent generations, and (2) any members of the same matrilineal groups through whom kinship must be reckoned are in equivalent generations unless this directly conflicts with principle 1.

It follows that the denotata of all lexemes in Group 1 (§4.1) are in senior generations, those of all lexemes in the first subset of Group 2 (§4.21) are in ego's generation, and those of the lexeme in Group 3 (§4.3) are in junior generations.

STRUCTURAL ASPECTS OF SIGNIFICATION

6. Components of signification. In stating the criteria for each lexeme we state what it signifies, its significatum. Each significatum consists of a combination of values for several different criteria. The criteria of which these are specific values can be construed as variables; for example, 'male' is a value of the variable of sex. Just as the phonemes of a language enter into various combinations to make morphemes, and these combine to form constructions, conceptual variables and their values combine in larger constructions which are the significata of linguistic forms.

To see these conceptual structures more clearly, let us translate the Trukese kinship terminology into a notation which will represent not the phonological elements of the speech symbols, but the conceptual components of their significata.

Let the letter A represent the constant of being *tefej* to ego, and the letters B–J the following variables, with values as indicated:

B, seniority of generation, with the values B_1 senior, B_2 same, and B_3 junior—generation to be understood according to the principles stated in §5;

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C, sex of the relative, with C_1 male and C_2 female;

D, symmetry or parallelism of relationship to the connecting matrilineal group, with D_1 symmetrical and D_2 asymmetrical;

E, sex relative to ego's sex, with E_1 same sex and E_2 opposite sex;

F, mode of relationship, with F_1 consanguineal and F_2 affinal;

G, age relative to ego's age, with G_1 older and G_2 younger;

H, matrilineal group membership relative to ego's, with H_1 member of ego's group, H_2 member of ego's father's group, and H_3 member of neither group;

J, collateral removal, with J_1 lineal and J_2 not lineal.

With this notation we now write the significatum of each kinship lexeme, as shown in Paradigms 1 and 2. Lexemes appear at the left, values of significata at the right.

PARADIGM 1

semej	AB_1C_1
semenapej	$AB_1C_1J_1$
no lexeme	$AB_1C_1J_2$
jinej	AB_1C_2
jinenapej	$AB_1C_2J_1$
no lexeme	$AB_1C_2J_2$
pwiij	$AB_2D_1E_1$
mwëgejej	$AB_2D_1E_2F_1$
mwääni	$AB_2D_1E_2F_1C_1$
feefinej	$AB_2D_1E_2F_1C_2$
no lexeme	$AB_2D_1E_2F_2$
jëësej	$AB_2D_2E_1$
pwynywej	$AB_2D_2E_2$
neji	AB_3

PARADIGM 2

jääj mwään	$AH_1E_1G_1$
mwääninyki	$AH_1E_1G_2$
no lexeme	AH_1E_2
jinejisemej	AH_2C_2
no lexeme	AH_2C_1
no lexeme	AH_3

6.1. Sememes and lexemes. Our analytical notation makes it possible to examine the utility of Bloomfield's definition of a sememe as the meaning of a morpheme.[17]

If we take a sample of the linguistic contexts in which a morpheme can occur, and then get a sample of its possible denotata for each context, we can establish a tentative significatum for each context. Then, by comparing the significata and denotata for each context with each other as well as with complementary denotata for other linguistic forms in similar contexts, we usually find that we can group the significata for at least several of these contexts together under one significatum, perhaps ending with a single significatum for the morpheme, perhaps with several significata. When we find contrasting significata for a morpheme in the same linguistic context, we are forced to decide that we are dealing with homonyms (*The night bore down upon them*, *The knight bore down upon them*). It frequently happens, however, that while we do not find contrasting significata in the same context, we find them in different contexts: the several significata are

[17] *Language* 162; see also Floyd G. Lounsbury, *Oneida verb morphology* 11 (*Yale Univ. publ. in anthr.*, No. 48, 1953), and Nida, op. cit.

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in complementary distribution with respect to the linguistic contexts in which they occur. Since they are in this sense noncontrasting, it is tempting to regard them as aspects of a single over-all meaning. This over-all meaning of a morpheme is presumably what Bloomfield meant by a sememe: a collection of significata in complementary distribution with respect to linguistic contexts. In such a view, each significatum would then be an alloseme. In contrast with this usage, we have used sememe as a synonym for significatum and alloseme for a denotative type. (Analogically a seme by our usage is a denotatum.) As we are using the term, a morpheme may be expected to have more than one sememe.

To make clear the reason for our departure from Bloomfield's usage, let us consider the Trukese morphemes **feefina* (with allomorphs *feefin*, *feefine/a/o-*, and *-feefin*)[18] and **mwääni* (with allomorphs *mwään*, *mwääni/u-*, and *-mwään*).

In one set of linguistic contexts, the allomorphs *feefin* and *mwään* consistently have denotata which are all adult female humans and adult male humans respectively. If we let X equal the value 'human' and Y a variable of age (in which Y_1 is 'adult,' Y_2 'adolescent,' and Y_3 'child') we find the significata of *feefin* in these contexts to be XY_1C_2 and those of *mwään* to be XY_1C_1. Other morphemes have allomorphs which in similar contexts are their denotative complements: *fëëpwyn* (XY_2C_2), *jënywën* (XY_2C_1), *neggin* (XY_3C_2), and *jäät* (XY_3C_1). All of these forms together with *feefin* and *mwään* clearly belong to a single paradigm.

The allomorphs *feefina/e/o-* and *mwääni/u-*, when coupled with suffixed possessive pronouns or the possessive linking form *-n*, invariably denote kinsmen of the possessor, having significata which we have written in Paradigm 1 as $AB_2D_1E_2F_1C_2$ and $AB_2D_1E_2F_1C_1$. These significata are not predictable from those of *mwään* (XY_1C_1) and *feefin* (XY_1C_2), since suffixed possessive pronouns do not otherwise signify kin relationships, nor can these two sets of significata conceivably belong to the same paradigms, not being complementary with respect to their denotata, although the forms which signify them are complementarily distributed with respect to their linguistic environments. The presence of the components C_1 and C_2 in the significata of *mwääni* and *feefinej* relates them semantically to *mwään* and *feefin*, but only in the same manner as it relates them to *semej* and *jinej* or to the phrases *neji mwään* and *neji feefin*.

In another set of linguistic contexts, with one exception, the allomorphs *feefin* and *mwään* simply indicate that whatever is denoted by a preceding form is a 'female human' or 'male human'; we would write their significata simply as XC_2 and XC_1. Thus the phrase *neji feefin* has the significatum AB_3XC_2 or, since X is already implied by A, AB_3C_2 (we have already seen that *neji* signifies AB_3). Similarly the phrase *neji mwään* signifies AB_3C_1.

The exception is *mwään* after an allomorph of the morpheme **jaa* when the latter is coupled with a possessive suffix, as in *jääj mwään*. Here we find a significatum $AH_1E_1G_1$, as written in Paradigm 2, in which the variable component C does not enter at all, although its derivation can still be traced through a connotational chain. We agree with Nida (op.cit. 9), who criticises the 'common

[18] The asterisk indicates a morpheme. Strictly speaking all these are allomorphs of a morpheme **fina* (as in *fine-n nöömw* and *fin acaw*); but to introduce the other allomorphs would complicate the presentation without affecting the point to be illustrated.

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denominator' approach to meaning on the grounds that 'in a series of contexts a word may have practically no common denominator, and yet the series exhibits obvious relationships.' The obvious relationships in this case, however, turn out to be no more than those which obtain between morphologically unrelated forms, e.g. the common component C_1 in the significata of *mwääni*, *semej*, *semenapej mwään*, *neji mwään*, *jäät*, and *jënywën*.

Since the several significata of a morpheme can be as unlike or as like each other as the significata of different morphemes, it serves no useful purpose to talk about 'the meaning of a morpheme'. It is more useful to define a sememe as the significatum of a lexeme. It is in this sense that we shall speak of a sememe from now on.

The foregoing examples also make it possible for us to refine our definition of a lexeme. We have already indicated that we regard the construction *jääj mwään* as a single lexeme because its significatum does not follow from the significata (in similar contexts) and arrangement of its parts. On the other hand, the syntactically identical construction *neji mwään* (AB_3XC_1) consists of two lexemes, *neji* (AB_3) and *mwään* (XC_1). We also regard as distinct lexemes morphologically identical forms which have different significata in different though complementary contexts. Thus *mwään* (XY_1C_1) is a different lexeme from *mwään* (XC_1). A lexeme, then, is a morpheme, construction, or phrase in those linguistic contexts where it has a single and unpredictable significatum. Its significatum is a sememe.

6.2. The componential structure of sememes. The conceptual variables and their values all relate to a consistent difference between two sets of phenomena as denotata of linguistic forms. Thus, the difference between the denotata of *semej* (AB_1C_1) and *jinej* (AB_1C_2) is a value of C (sex of the object), while the difference between *semej* and *neji* (AB_3) is a value of B (seniority of generation). The components of signification, then, are the formal criteria by which we differentiate one thing from another. Our notation for writing sememes, crude as it is in its present form, is a method for symbolizing concepts which is functionally and structurally equivalent to the phonemic method of symbolizing speech forms. Such concept forms as $AB_2D_1E_1$ (*pwiij*) are structurally analogous to morphemes. The structural phonological relationship of English *pet* and *bet*, for example, parallels the structural conceptual relationship of AB_1C_1 (*semej*) and AB_1C_2 (*jinej*).

According to the distinction customarily drawn between ideographic and phonemic writing, the written form AB_1C_2 is an ideogram; it stands for an 'idea' rather than for spoken sounds. From a functional point of view, however, any spoken word in a language is like an ideogram, because it, too, stands for an idea. The advantage of phonemic writing is that the structure of the written forms reflects the structure of the corresponding spoken forms. Our notation functions for concepts in the same way as phonemic notation for speech.

This is a fact of considerable importance. Because phonemic notation describes the structure of what it represents, it is possible by manipulating the notation to make observations of speech forms under conditions which an investigator can control. The rigor with which speech forms are isolated and

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analyzed by modern linguists is due more than anything else to the phonemic method of notation. I am convinced that further development of a notation for sememes will open the way for equally rigorous analysis of the content of conceptual systems, as far as linguistic and other behavior forms provide a pathway into them—and I know of no other pathway. Some idea of the nature of the potentialities of this notation are seen in connection with the structure of paradigms.

7. Paradigms. In my earlier account of Trukese kinship, I defined a semantic system as follows (*PKC* 107): 'We can say that a series of symbolic behavior patterns belong to the same semantic system if (1) their significata include one characteristic in common, (2) the differences between their significata are functions of one (simple system) or more (complex system) variable characteristics, and (3) their significata are mutually contrasting and complement each other.' Thus the properties of a semantic system are the same as those of a paradigm in traditional linguistic usage. In each case the significata of all of the lexemes or constructions are parts of a single conceptual universe—that of kinship in the present instance, that signified by the root or base form in a traditional paradigm.

That we have had to construct two paradigms instead of one for Trukese kinship follows from the fact that the significata of some lexemes are not complementary with those of the others. The former divide the universe in a way which cuts across the partitions of the same universe formed by the latter.

7.1. Paradigms as structures. Paradigms 1 and 2 both have definite structures based on the particular arrangements of their conceptual components. In the first paradigm, the only variable that is represented in all the significata is B (seniority of generation). Structurally, the concept of generation is basic to the system of meanings expressed by these lexemes. In the second paradigm it is variable H (membership in matrilineal groups) which is structurally the basic concept. (A partial reconciliation of B and H is reflected in the special definition of generation equivalence required for B.) The minor role played in kinship classification by variables C (sex of the relative), F (mode of relationship), and J (degree of collateral removal) is clearly portrayed in the paradigms. Our analysis has obvious advantages for the comparative study of conceptual systems like kinship in that it shows the structure of each system to be compared, revealing the concepts involved and their exact place within the whole. For kinship it now becomes possible to explore the kinds of structural differences to which Kroeber's insight so long ago called attention.[19]

7.2. Incomplete paradigms. Both paradigms reveal that there may be conceptual systems within a culture whose categories are not all represented by lexemes. The three lexemes in Paradigm 2 have significata which imply a minimum of three additional complementary forms. If we accept that it is impossible to have a concept 'male' without a concept 'female' (or at least 'not male'), a concept 'thin' without a concept 'thick', or a concept 'medium' without concepts for the extremes, the three concepts in Paradigm 2 for which there are no lexemes must nevertheless be a part of Trukese culture, active elements in Trukese thinking.

[19] *Journal of the Royal Anthropological Institute* 34.77–84 (1909).

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It is significant for the study of cultural forms that our analysis should enable us to get at concepts which are not lexically objectivized through those which are.

Every conceptual component, such as B_1 or C_2, could be represented by a lexeme or syntactic feature of some kind. So also could every one of the possible combinations of the several components, such as a theoretical combination $AB_1E_2H_1$.

The fact that only some of the many possible combinations in a given paradigm are symbolized by lexemes is not without cultural and psychological significance. To give them all symbolic expression in lexemes would, as Kroeber (ibid. 77) and after him Murdock[20] have pointed out, result in a superfluity of lexemes far beyond the number needed for practical problems of communication about social relationships. Why only certain conceptual variables are utilized, why only certain combinations of their values are symbolized in lexemes, are questions which have challenged many students of social organization;[21] but they are not our concern here.

7.3. Circumlocutions and semantic precision. I once said of Trukese kinship terminology (*PKC* 98) that 'it is quite possible for anyone to indicate the relationship between two people exactly by describing it, as we do in English'. It would have been better to say 'more precisely' instead of 'exactly'. There are, to be sure, linguistic forms in Trukese which when properly combined enable one to signify 'own sibling' as distinct from remoter kinsmen, i.e. *pwiij* (or *mwëgejej*) *emëccëk ineem me jemëccëk semeem*. To have identical *jinej* and *semej*, two persons must be siblings in our sense. To put together a combination of forms, however, which can signify only the conceptual category consisting of WiBrWi and HuSiHu ('zero lexeme' of Paradigm 1) would require at least a whole paragraph. The circumlocution that is regularly used, *pwynywen ëësej*, is ambiguous because it is also used to talk about anyone who is *pwynywej* to ego's WiBr or HuSi or to any of his other *jëësej*, obviously covering a lot of territory. The fact is that the significata of Truk's existing kinship lexemes are the complements of a conceptual category for which there is not only no lexeme but not even a practical combination of lexemes that unambiguously signifies it. We shall see shortly why WiBrWi and HuSiHu have not been included among the possible denotata of *mwëgejej* (are not allosemes of its sememe), as considerations of symmetry in Paradigm 1 would lead us to expect. The point remains that, while the vocabulary of a language cannot be expected to deal readily with concepts which do not exist for its speakers, it may also be unable to be precise about concepts which very clearly do exist.

7.4. Latent concepts and latent structures. Though they are not represented by lexemes, conceptual categories like WiBrWi and HuSiHu are a definite part of Trukese culture. So, too, are the conceptual components whose combinations formed the sememes or significata presented in Paradigms 1 and 2. Any concept which completes a paradigm or marks the consistent difference between the de-

[20] *Social structure* 96–7 (1949).

[21] For recent theories and a review of earlier ones, see Murdock, *Social structure* 113–83.

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notative sets of a pair of lexemes is an active element in the cultural structure. Having work to do, such a concept is not latent in the sense to be discussed here.

We noted above that our paradigms do not contain every possible combination of every one of the conceptual components represented in them. A theoretically possible combination such as $AB_1E_2H_1$ is not an active part of the Trukese conceptual repertoire. But such combinations are latent in Trukese culture, because all the necessary components are represented in other combinations forming active concepts. Should $AB_1E_2H_1$ ever be activated as a category of Trukese thought, it will represent a change in culture less basic than the development of an entirely new conceptual component, such as a variable of degree of generation removal (as distinct from generation seniority) or of similarity of ego's sex with the sex of the immediate relative through whom kinship is traced (technically known as the criterion of bifurcation; see Murdock 104).

The possibility of extending the use of a conceptual variable like J (degree of collateral removal) in combinations where it does not now occur is significant for the potentialities of Truk's present system of kinship classification for future change. Such an extension would not only bring to the fore what are as yet no more than latent concept combinations, it would also alter the structure of the existing paradigms and the conceptual systems which they represent. We may say, then, that any new structures which are possible through new combinations of components already present are latent in the culture.

7.5. Alternative concepts and structures. When two lexemes are such that any possible denotatum of either one can be a denotatum of the other, the lexemes are synonyms and have the same significatum or sememe. We now consider the possibility that two different sememes may be such that any possible alloseme of one may also be an alloseme of the other, so that they are effectively symbolized by the same lexeme, having in relation to it the same possible denotata. Two speakers who each have a different one of two such sememes in their conceptual repertoire will understand each other perfectly, because whichever sememe they refer to, the result will be the same. This possibility is realized in Trukese kinship terminology.

The lexemes *mwääni* and *feefinej* were represented in Paradigm 1 with the values $AB_2D_1E_2F_1C_1$ and $AB_2D_1E_2F_1C_2$ respectively. In §2.21, however, we saw that the difference between the two sets of denotata possible for this pair of lexemes can be equally accounted for either by variable C (sex of the relative) or by a variable of the sex of ego. If we set K for the latter (K_1 being male and K_2 female), it is evident that $AB_2D_1E_2F_1K_2$ and $AB_2D_1E_2F_1K_1$ refer to the same denotata as $AB_2D_1E_2F_1C_1$ and $AB_2D_1E_2F_1C_2$. Since C was already needed to describe the difference between the denotata of *jinej* and *semej*, we chose to describe the difference between *mwääni* and *feefinej* similarly. That either C or K alone is sufficient to account for the difference between these two, however, does not alter the fact that both the sex of the relative and that of ego must be known in order to use these lexemes correctly. Knowledge of both is implicit in variable E (sex of the kinsman relative to ego's sex). It is clear that K is as active a concept in Trukese culture as C.

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This being the case, why postulate a variable E? As illustrated in Paradigm 3, we can describe everything that E accounts for by the combined use of C and K. The obvious disadvantage, however, is that every lexeme in the paradigm except *mwääni* and *feefinej* thereby comes to have not one but two different sememes; instead of single lexemes they all become pairs of homonymous lexemes. While Paradigm 3 is theoretically adequate, it lacks the elegance of Paradigm 1. Nevertheless it is a model from which it is possible to infer the use of the lexemes cor-

PARADIGM 3

pwiij	$AB_2D_1C_1K_1$	or $AB_2D_1C_2K_2$
mwëgejej	$AB_2D_1C_1K_2F_1$	or $AB_2D_1C_2K_1F_1$
mwääni	$AB_2D_1C_1K_2F_1$	
feefinej		$AB_2D_1C_2K_1F_1$
no lexeme	$AB_2D_1C_1K_2F_2$	or $AB_2D_1C_2K_1F_2$
jëësej	$AB_2D_2C_1K_1$	or $AB_2D_2C_2K_2$
pwynywej	$AB_2D_2C_1K_2$	or $AB_2D_2C_2K_1$

PARADIGM 4

pwiij	$AB_2E_1L_1$
mwëgejej	$AB_2E_2L_1$
jëësej	$AB_2E_1L_2$
pwynywej	$AB_2E_2L_2$
pwiij	$AB_2E_1L_3$
no lexeme	$AB_2E_2L_3$

rectly. If one person has in mind the conceptual scheme of Paradigm 1 and another the scheme of Paradigm 3, they will understand each other just as well as if both used the same model.

To take another example, we have seen that the difference between the two sets of possible denotata for *pwiij* ($AB_2D_1E_1$) and *jëësej* ($AB_2D_2E_1$) is best expressed by D (parallel connection to the intervening matrilineal group). The fact remains, however, that all denotata of *jëësej* are kinsmen by marriage only. We did not use the variable F (mode of relationship) to describe the difference between these two denotative sets because some denotata of *pwiij* are also affinal kinsmen. In the latter case, however, the affinal kinsmen are all persons married into one's spouse's group, persons whose connection to ego involves two intervening marital links as distinct from only one. If we set L for a variable of degree of affinal removal with the values L_1 zero degree of removal, i.e. consanguineal, L_2 one degree of removal, and L_3 two degrees of removal, we get the conceptual system shown in Paradigm 4 for persons in ego's generation. There is much to commend this interpretation of the significata of these lexemes[22] as against the one shown in Paradigm 1. Here we can describe the system with one less variable, L doing the work of both D and F. The disadvantage is that, whereas before we could group all denotata of *pwiij* in a single category and give *pwiij* a single sememe, we must here treat *pwiij* as two homonymous lexemes with two distinct sememes. We choose at the moment to favor the interpretation which reduces homonyms, but grant that there may be occasions when the interests of simplicity are better served by keeping homonyms and thereby reducing the number of conceptual variables. The large number of homonyms obtained in Paradigm 3, on the other hand, is clear evidence that the interpretation which it represents is undesirable for descriptive purposes.

Regardless of what is desirable from a descriptive standpoint, Paradigms 1, 3, and 4 are all valid, in that they all enable us to predict what can and what

[22] This was, in fact, the view published in *PKC* 107–8.

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cannot be a denotatum for each of the lexemes therein. Since we can reproduce Trukese kinship usage equally well from any one of these three paradigms, we cannot eliminate from the Trukese cognitive world any one of the alternative conceptual variables E, K, D, F, and L. If there is a future change in Trukese kinship usage by which L, for example, becomes the only criterion (as against D and F) to account for the difference between the denotative sets for the different lexemes, such a change will represent no more than the resolution of conceptual alternatives already present in Trukese culture in favor of one of them.

7.6. Asymmetry of paradigms. It is possible to speak of paradigms as symmetrical or asymmetrical in structure, depending on the consistency with which the conceptual components combine with one another throughout the paradigm. Thus Paradigm 4 is symmetrical in a way that Paradigm 1 is not. In Paradigm 1, moreover, the presence of a lexeme for AB_3 (kinsman of junior generation regardless of sex) is not balanced by a reciprocal lexeme for AB_1 (kinsman of senior generation regardless of sex). Similarly, in ego's generation the remaining complementary category $AB_2D_1E_2$, implied by the sememes $AB_2D_1E_1$ (*pwiij*) $AB_2D_2E_1$ (*jëësej*), and $AB_2D_2E_2$ (*pwynywej*), is not signified by a lexeme, but is subdivided into two categories, $AB_2D_1E_2F_1$ and $AB_2D_1E_2F_2$, the former signified by *mwëgejej* and the latter by no lexeme, being only ambiguously signified by the circumlocution *pwynywen ëësej* (see §7.3). Are such breaks in otherwise symmetrical patterns fortuitous, or have they functional significance?

It is generally assumed by anthropological students of social organization today that there is close (if not perfect) congruence of the distribution of different kinds of role relationship among kinsmen with the grouping of these kinsmen into categories signified by the several kinship terms.[23] While there is a tendency to oversimplify the nature of this relationship and to assume, perhaps, a higher degree of near congruence than actually obtains, some kind of functional relationship between social roles and the terminology of social classification can be regarded as axiomatic. There are probably a number of different kinds of such functional relationship. An interesting one of a sort which has not been observed hitherto appears to be responsible for the asymmetry of Paradigm 1 in classifying kin of ego's generation.[24]

The question here is why the term *mwëgejej* is not extended, like *pwiij*, to include persons married into one's spouse's matrilineal kin groups (affines with whom one has a parallel connection) and why the circumlocution *pwynywen ëësej* (or 'zero lexeme') appears at this point to break the over-all pattern of the paradigm. Figure 2 illustrates a group of men (M) and women (W) who are consanguineally related and in the same generation, together with their wives (w) and husbands (m). The members of each symmetrically related pair of the same sex (MM, WW, mm, ww) refer to each other as *pwiij*, the members of each asymmetrically related pair of the same sex (Mm, Ww) refer to each other as *jëësej*, and the members of each asymmetrically related pair of opposite sex (Mw, mW) refer to each other as *pwynywej*. The problem is that we should expect by analogy that the members of every symmetrical pair of opposite sex (MW,

[23] Murdock 106–12.

[24] I am indebted to John Cole for first calling this to my attention.

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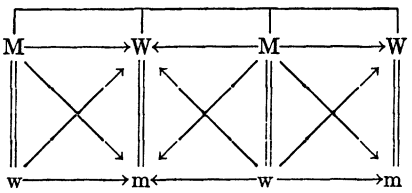


FIGURE 2

mw) would refer to each other as *mwëgejej*, whereas only those in the pair MW do so, while persons of the pair mw refer to one another as *pwynywen ëësej*.

If we turn to the rules of behavior in kinship we find that all M in Figure 2 have authority over all W (as shown by the arrows). The rule for extending behavioral roles to one's in-laws after marriage is that one has the same behavioral obligations to one's spouse's relatives as one's spouse. By this rule all M have authority over all m, all w over all W, and all w over all m. In the last instance the flow of authority between the sexes is opposite to that between M and W. For m and w to use the same kinship terms for each other as do M and W would make for a striking inconsistency in the behavioral connotations of the terms. The pattern for the extension of behavioral roles to in-laws is in direct conflict with the pattern of kinship terminology at this point. Something has to give way to resolve the conflict; it is the terminological pattern which yields to the behavioral, the significational system to the connotational. Since, on the other hand, there is between m and m the same behavioral relationship (relating to authority) as between W and W, and the same relationship between w and w as between M and M, there is here no conflict between the terminological pattern and that for extending behavioral roles.

The appearance of the 'zero lexeme' and consequent circumlocution *pwynywen ëësej*, in violation of the otherwise symmetrical pattern of the paradigm, does not reflect an important special relationship to be differentiated from all others (in-laws of one's in-laws are not that important); but it does reflect a point where two intersecting systems with independent structures come together in a way that is contradictory as compared with the pattern of their intersection at other points. The function of the 'zero lexeme' seems to be to resolve a psychologically disturbing contradiction or logical impasse. The fact that no new lexeme has been coined may be attributed to the compelling force of the analogs within the pattern of signification itself. There is, apparently, no new term for HuSiHu and WiBrWi because these kin types are still felt to be inappropriate as anything but *mwëgejej*, even though the connotations of the term effectively inhibit its use.

CONCLUSIONS

8. Applicability of the method. Since we have confined our discussions to the limited universe of Trukese kinship, the question arises to what extent the method here proposed is applicable to other universes. Do lexemes pertaining to other subjects possess complementary sememes forming paradigms? Certainly the great majority of them must do so. Direction terms, for example, form

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obvious paradigms, as do the host of lexemes which signify such personal attributes as sex, age, complexion, character traits, body builds, and so on. A method based on determining the consistent difference between the possible denotata or other contextual aspects of linguistic forms, therefore, should be applicable to the problem of empirically deriving their significata.

Because we have dealt with kinship terminology, however, we have not illustrated the range of conceptual forms or types of paradigm structures which this method is likely to produce. In some universes the complementary categories are not perfectly discrete in the sense that no denotatum of one lexeme can be a denotatum of another. Color terms are a case in point. While an informant may insist that *red* and *pink* are different colors which are not to be confused, there will be some denotata possible for one which are also possible for the other. Since such confusion occurs only at the boundaries and is due to the relativity of perceptual contrasts, it does not necessarily contradict the complementary nature of these categories at the conceptual level.[24a]

Color terms call attention to another problem as well. The components of the significata of such terms are impossible to isolate and describe without resort to the language of spectrum analysis. The only way that I can otherwise make it possible for someone else to form a concept of what it is the Trukese mean by their color terms *jarawaraw*, *parapar*, *jönojön*, etc., is to present him with concrete examples covering the range of actual colors which can be denoted by each term. While the terms clearly belong together in one paradigm as referring to complementary categories of color, their significata elude precise definition. The necessity to conceptualize differences between things which the analyst has never thought about before, and to discover adequate theories to account for them, poses a tremendous challenge. The Trukese concept of generation presented some difficulties of this kind. By persevering with the method, however, it is often possible, as in this case, to work the problem through, and as a result to find one's understanding of cultural phenomena greatly increased.

9. Utterances of more than one lexeme. An aspect of signification that we have not touched on at all is the process of combining sememes when lexemes are put together in complete utterances. In defining the lexeme as we did, we assumed that the signification of a complete utterance can be derived from the sememes and syntactic arrangement of its lexemes. But just as morphemes are likely to undergo various modifications in shape depending on their position in an utterance, sememes probably undergo comparable modifications. Thus an allomorph may have different shades of meaning as head and as modifier in a noun phrase, as shown in the case of *mwään* in §6.1. It may be possible to set up sememe-lexeme classes with all members of the same class showing the same type of modification according as their function in the utterance changes. Once we are in a position to write whole utterances in a notation symbolizing sememes in terms of their conceptual components, along the lines suggested here, we shall undoubtedly discover many things about the 'syntax' of conceptual processes.

[24a] See, for example, Harold C. Conklin's analysis of Hanunoo color categories, *Southwestern journal of anthropology* 11.339–44 (1955).

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10. Signification and behavior. Our analyses reveal the necessity of carefully distinguishing those things which tend to be associated with a particular category of kin from those things which are criteria for being in the category and therefore define it. The behavior which one ideally exhibits to persons in a given kinship category is obviously not a criterion of the category but something associated with it. Behavior patterns which properly obtain in my relations with my brother, for example, are connoted by the expression *my brother* but are not a part of what it signifies, because I know that so-and-so is my brother whether he acts like it or not. This does not mean, on the other hand, that the significata of all labels for types of persons exclude behavioral criteria. In order to be a *football hero*, one must have behaved in certain ways; but one need not behave in any particular way to qualify as someone's *cousin*. If a man is or is not *President of the United States* depending on the electoral behavior of others, is or is not a *boor* depending on his own behavior in the company of others, and is or is not a *youth* irrespective of anyone's behavior, there is clearly no simple relationship between linguistic forms and other forms of behavior.[25]

Another dimension of meaning and behavior not touched on in this paper is the role of linguistic utterances in social interaction as gestures. Expressions like *Hello!*, *Get out of here!*, *Won't you sit down?*, *How are you?*, *It's a nice day*, *Come here!*, and *Please* (Truk has its counterparts) are signs of status relationships (you don't say *Come here!* to everybody) and expressions of status obligations, rights, privileges, or powers. Their analysis must be conducted along somewhat different lines from those presented here.[26]

Finally, in suggesting a way in which methods in linguistic analysis may be applied to descriptive semantics, I have sought to avoid entanglement in general semantic theory. Adequate theory can develop, it seems to me, only as we seek seriously to describe real systems of meaning as manifest in the contexts of linguistic utterances. There can be no other successful approach than the one outlined by Kroeber in describing his own work:[27]

> My natural and first interest always has been in phenomena and their ordering ... From the ordering, general conclusions emerge; and, with these, eventuate certain principles as to how best to arrive at valid conclusions; in one sense theory therefore consists for me in considerable part of methodology. Perhaps the foregoing will make clear why I speak of theory as a by-product. It is not in a belittling sense. The theory just was sweated out piecemeal and slowly over fifty years.

[25] See my comments in *Lg.* 31.243, and my discussion of kinship behavior as related to kinship terms in *PKC* 115–8.

[26] For examples of such analyses, see *PKC* 111–9.

[27] Alfred L. Kroeber, *The nature of culture* 3 (1952).

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